

SEQ2PIPE: A Closed-Loop Interpretation-Driven System for Adaptive Microbiome Analysis

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<https://github.com/Rhizobium-gits/seq2pipe>

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Abstract

Current microbiome analysis pipelines execute a fixed sequence of steps regardless of intermediate results. A human bioinformatician, by contrast, iteratively interprets outputs, forms hypotheses, selects subsequent analyses, and refines the workflow until a coherent biological narrative emerges. We present SEQ2PIPE, a closed-loop system that reproduces this interpretation–decision–execution cycle computationally. The system consists of three layers: (1) a **DataInspector** that directly parses QIIME2 exports and executes nonparametric statistical tests in pure Python, (2) an **EvalMediator** that scores each analysis step across eight quantitative axes and compresses the accumulated knowledge state via truncated SVD, and (3) a **DecisionEngine** that applies literature-grounded decision trees to deterministically select, skip, or add subsequent analyses. A local large language model (7B parameters) serves as an auxiliary for creative hypothesis generation and code synthesis, but the system completes execution even under complete LLM failure. Evaluation on two public datasets (Moving Pictures, 34 samples; Gravity experiment, 138 samples) shows that the DecisionEngine correctly adapts: adding all beta follow-ups when PERMANOVA is significant ($p = 0.005$) and skipping 4 steps when it is not ($p = 0.81$). Registry analysis steps achieve 54% success, while LLM-generated investigations achieve 6%, revealing current limitations of 7B-parameter models for autonomous code generation. We discuss convergence behavior, reproducibility, and limitations, and argue that interpretation-driven adaptive systems represent a necessary evolution beyond static pipelines.

1 Introduction

Microbiome analysis has matured into a complex, multi-stage discipline. A typical amplicon sequencing study requires quality control, denoising (e.g., DADA2 [Callahan et al., 2016]), phylogenetic tree construction, alpha and beta diversity analysis, taxonomic classification, differential abundance testing, and downstream visualization—each with parameter choices that depend on the data at hand.

The rigidity problem. Current pipeline frameworks—QIIME2 [Bolyen et al., 2019], Snake-make [Köster & Rahmann, 2012], Nextflow [Di Tommaso et al., 2017]—encode a *fixed* directed acyclic graph (DAG) of steps. Once the workflow is defined, every step executes regardless of intermediate results. If PERMANOVA yields $p = 0.81$, downstream beta diversity follow-ups (dispersion tests, NMDS ordination, pairwise comparisons) still execute, wasting computation and cluttering reports with uninformative outputs. Conversely, an unexpected finding (e.g., a 379-fold enrichment of *Escherichia-Shigella*) triggers no automatic follow-up; the analyst must manually intervene.

How humans analyze data. An experienced bioinformatician operates in a closed loop:

1. **Interpret:** inspect the output of the current step (“Shannon $p = 0.034$, but effect size is small”).
2. **Hypothesize:** form a conjecture (“This may be a sampling artifact”).
3. **Select:** choose the next analysis to test the hypothesis (“Run rarefaction curves to check depth dependence”).
4. **Refine:** update the mental model and repeat.

This loop continues until the analyst reaches a stable, internally consistent interpretation. No existing pipeline system reproduces this behavior.

Contribution. We introduce SEQ2PIPE, a system that computationally reproduces the human analytical reasoning loop. Our key contributions are:

- A **three-layer architecture** that separates data inspection (deterministic), evaluation and decision (deterministic), and code generation (LLM-assisted), yielding an **LLM-failure-tolerant** adaptive system.
- A **literature-grounded decision engine** that encodes established analytical heuristics (Anderson, 2001; Willis, 2019; Gloor et al., 2017; McMurdie & Holmes, 2014) as deterministic decision trees, enabling reproducible, auditable analytical decisions.
- An **eight-axis quantitative scoring framework** that evaluates each analysis step from raw data—not from LLM text output—providing a grounded basis for adaptive decisions.

2 Related Work

Static pipeline frameworks. QIIME2 [Bolyen et al., 2019] provides a plugin-based architecture with provenance tracking but no conditional branching based on intermediate results. Snakemake [Köster & Rahmann, 2012] and Nextflow [Di Tommaso et al., 2017] support conditional rules, but the conditions must be specified *a priori* by the user; the system does not inspect result quality or statistical significance to alter the workflow.

AutoML and automated statistics. Auto-WEKA [Kothari et al., 2019] and Auto-sklearn [Feurer et al., 2015] automate model selection via Bayesian optimization but target supervised learning, not the open-ended exploratory analysis characteristic of microbiome studies. Statistical analysis planning tools such as Wasserstein et al. [2019] provide decision trees for test selection but do not execute analyses or iterate on results.

LLM-based analysis assistants. Recent systems such as Data Interpreter [Hong et al., 2024] and ChatGPT Advanced Data Analysis use LLMs to generate and execute code. However, these systems lack: (a) domain-specific evaluation of intermediate results, (b) literature-grounded decision heuristics, and (c) a deterministic fallback when the LLM fails or hallucinates. SEQ2PIPE addresses all three gaps.

3 Conceptual Framework

We formalize the analytical reasoning loop as follows.

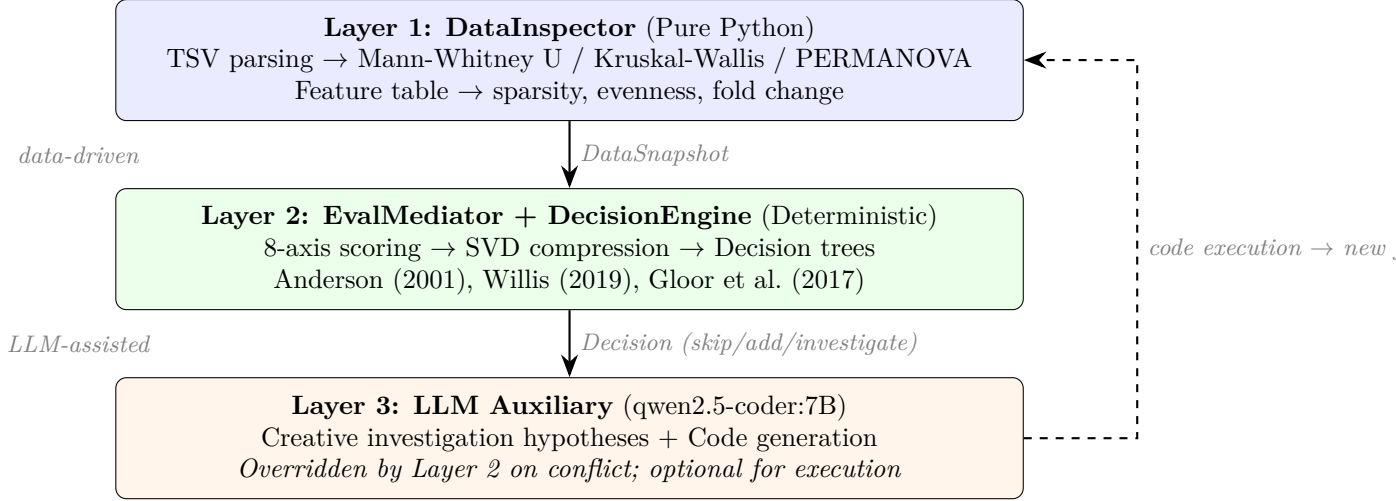


Figure 1: Three-layer architecture of SEQ2PIPE. Layers 1 and 2 are fully deterministic and LLM-independent. Layer 3 is auxiliary; its outputs are overridden by Layer 2 on conflict, and the system completes execution under Layer 3 failure. Dashed arrow indicates the feedback loop.

Definition 1 (Analytical State). *At iteration t , the analytical state S_t is a tuple (D, H_t, P_t, R_t) where D is the immutable dataset, H_t is the accumulated set of findings (hypotheses confirmed or rejected), P_t is the current analysis plan (an ordered list of pending steps), and R_t is the set of completed results.*

Definition 2 (Closed-Loop Analytical System). *A system is closed-loop if, after executing step $s_t \in P_t$ and observing result r_t , it updates:*

$$H_{t+1} = H_t \cup \text{interpret}(r_t) \quad (1)$$

$$P_{t+1} = \text{decide}(H_{t+1}, P_t, D) \quad (2)$$

where *interpret* extracts quantitative findings from r_t (not from textual summaries) and *decide* modifies the plan by adding, removing, or reordering steps.

Definition 3 (Convergence). *The system converges if there exists T such that for all $t \geq T$, $P_t = \emptyset$ (the plan is exhausted) and the execution error count $e_t = 0$. We define the convergence depth as the total number of decision-execution cycles before termination.*

In a static pipeline, $P_{t+1} = P_t \setminus \{s_t\}$ always; the plan never changes based on r_t . SEQ2PIPE implements a non-trivial decide function that depends on quantitative properties of H_{t+1} and D .

4 System Architecture

4.1 Layer 1: DataInspector

The `DataInspector` class directly parses QIIME2 exported TSV files without relying on LLM text output or stdout pattern matching. It implements:

Alpha diversity testing. For each alpha metric TSV (Shannon, observed features, Pielou’s evenness), sample values are read and partitioned by group using a metadata-derived group

Table 1: Eight evaluation axes with default weights and computation methods.

Axis	Weight	Computation
statistical_significance	0.20	$\min(1, -\log_{10}(\min p)/4)$ from alpha/beta tests
biological_relevance	0.18	$\min(1, \text{fc}_{\max}/10 + \Delta_{\max}/20) + \text{known taxa bonus}$
actionability	0.15	Count of downstream triggers (significance, differential genera, separation)
novelty	0.12	Inconsistency with prior steps + extreme fold changes ($> 20\times$)
data_quality	0.10	Base 0.7, penalized by sparsity (> 0.9 : -0.2), CV (> 0.5 : -0.1)
effect_magnitude	0.10	$\max(\text{Cliff's } d, F/3, \text{fc}/10)$
confidence	0.08	Step function on minimum group size (≥ 20 : 0.9 , ≥ 10 : 0.7 , $\dots$)
reproducibility	0.07	Consistency of significance direction within analysis category

map. For two groups, a Mann-Whitney U test is computed via rank transformation with tie correction and normal approximation:

$$z = \frac{U - n_1 n_2 / 2}{\sqrt{\frac{n_1 n_2}{12} \left(N + 1 - \frac{\sum_j (t_j^3 - t_j)}{N(N-1)} \right)}} \quad (3)$$

where t_j are tie group sizes and $N = n_1 + n_2$. For three or more groups, a Kruskal-Wallis H test is used:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1) \quad (4)$$

with tie correction and χ^2 approximation via the Wilson-Hilferty transformation [Wilson & Hilferty, 1931]. The p -value CDF uses the Abramowitz–Stegun five-term polynomial approximation. Cliff’s delta and η^2 are computed as nonparametric effect sizes.

Beta diversity testing. Distance matrices are read in full ($O(n^2)$ entries). Within-group and between-group distances are separated. A pseudo-PERMANOVA [Anderson, 2001] F -statistic is computed as:

$$F = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(N-k)} \quad (5)$$

with p -value estimated from 199 permutations (fixed seed for reproducibility). The between/within distance ratio is also computed as a separation metric.

Taxonomic differential analysis. The feature table and taxonomy file are joined. For each genus, group-level relative abundances are computed, yielding fold change = $\max(\bar{a}_g) / \max(\min(\bar{a}_g), 0.01)$ and absolute difference. Results are sorted by absolute difference.

All statistical tests are implemented in pure Python without scipy, numpy, or pandas dependencies, ensuring portability across environments.

4.2 Layer 2: EvalMediator and DecisionEngine

Eight-axis scoring. Each completed analysis step is scored on eight axes (Table 1) using the DataSnapshot from Layer 1, not stdout text.

The composite score is the weighted sum $c_t = \sum_a w_a \cdot s_{t,a}$. Weights are adjustable per experiment type (e.g., antibiotic studies increase biological_relevance to 0.22).

SVD compression. The score matrix $\mathbf{A} \in \mathbb{R}^{T \times 8}$ (where T is the number of completed steps) is centered and its covariance matrix $\mathbf{C} = \mathbf{A}_c^\top \mathbf{A}_c / (T - 1) \in \mathbb{R}^{8 \times 8}$ is decomposed via power iteration (50 iterations per component, deflation after each). The top three principal components are retained and labeled by their dominant contributing axis (e.g., “statistical certainty” if the largest loading is on significance). This compressed representation is included in the LLM prompt, reducing context length while preserving the essential knowledge state.

DecisionEngine. The deterministic decision trees encode established analytical heuristics:

- **Alpha branch** (Willis, 2019): If $\min p_\alpha < 0.05$ and effect size > 0.5 : add effect size plot and rarefaction. If $\min p_\alpha < 0.05$ but effect size ≤ 0.5 : add rarefaction only (sampling artifact check). If $\min p_\alpha \geq 0.05$: skip alpha follow-ups.
- **Beta branch** (Anderson, 2001): If $\min p_\beta < 0.05$: add betadisper (mandatory). If separation ratio > 2.0 : add NMDS and pairwise PERMANOVA. If $\min p_\beta \geq 0.05$: skip all beta follow-ups (dispersion, NMDS, t-SNE, heatmap).
- **Taxonomy branch** (Gloor et al., 2017): If fold change > 50 at abundance $< 1\%$: flag as compositional artifact, add filtered retest. If fold change > 5 : add differential abundance test. If ≥ 5 genera with $> 2\times$ change: add functional prediction (PICRUSt2) and indicator species analysis.
- **Quality branch** (McMurdie & Holmes, 2014): If read depth CV > 0.5 : add rarefaction curves. If sparsity $> 90\%$: add prevalence-filtered reanalysis.

Merge with LLM. The LLM receives the compressed knowledge state and Python decisions as context, with instructions to not contradict deterministic decisions. The `merge_with_llm()` function combines outputs: Python-determined skips override LLM-proposed additions; LLM-proposed investigations are appended if they do not conflict with Python skips.

4.3 Layer 3: LLM Auxiliary

A local 7B-parameter model (qwen2.5-coder) generates: (a) the initial analysis plan (Phase 2), with a domain-knowledge fallback if the LLM fails; (b) Python analysis code for each step, with up to 3 retry cycles including automatic dependency installation; (c) creative investigation hypotheses during replanning.

Critically, **the system completes execution under total LLM failure**: Phase 2 falls back to `build_domain_driven_plan()`; Phase 3 replanning applies Python decisions only; individual step failures are recorded and skipped.

4.4 Execution Loop

Algorithm 1 formalizes the adaptive execution loop.

5 Key Innovation: What seq2pipe Is Not

It is important to delineate what SEQ2PIPE is *not*:

- **Not a static pipeline.** Unlike QIIME2 or Snakemake, the execution plan changes after each step based on quantitative evaluation of results (Figure 2).
- **Not an AutoML system.** AutoML optimizes a fixed objective (e.g., classification accuracy) via search. SEQ2PIPE has no single objective; it navigates an open-ended analytical space where “what to analyze next” depends on what was found so far.

Algorithm 1 Adaptive Execution Loop

Require: Dataset D , metadata M , research question Q , plan P_0

Ensure: Analysis results R , score history $\mathbf{A}$

```
1: mediator  $\leftarrow$  EvalMediator( $D, M$ )
2:  $t \leftarrow 0$ 
3: while  $P_t \neq \emptyset$  do
4:    $s_t \leftarrow P_t.\text{pop}()$ 
5:   code  $\leftarrow$  LLM.generate( $s_t, Q, D$ )
6:   (ok, stdout, figs)  $\leftarrow$  execute(code)
7:   if  $\neg \text{ok}$  and retries  $< 3$  then
8:     code  $\leftarrow$  LLM.fix(code, stderr); goto 6
9:   end if
10:  snap  $\leftarrow$  mediator.inspector.inspect( $s_t, \text{figs}$ ) {Layer 1}
11:  score $_t \leftarrow$  mediator.evaluate(snap) {Layer 2: 8-axis scoring}
12:  if  $t \bmod 4 = 0$  or score $_t$ .novelty  $> 0.7$  then
13:     $d_{\text{py}} \leftarrow$  DecisionEngine.decide( $P_t$ ) {Layer 2: deterministic}
14:     $d_{\text{llm}} \leftarrow$  LLM.replan(state,  $P_t$ ) {Layer 3: optional}
15:     $P_{t+1} \leftarrow$  merge( $d_{\text{py}}, d_{\text{llm}}, P_t$ ) {Python wins on conflict}
16:  else
17:     $P_{t+1} \leftarrow P_t$ 
18:  end if
19:   $t \leftarrow t + 1$ 
20: end while
```

- **Not an LLM assistant.** Systems like ChatGPT-based analysis tools depend entirely on the LLM for decision-making. SEQ2PIPE’s decisions are primarily deterministic (Layers 1–2); the LLM is auxiliary and expendable. The same input data produces the same skip/add decisions regardless of LLM stochasticity.

The core novelty is the combination of *data-grounded evaluation* (not text parsing), *literature-based deterministic decisions* (not LLM judgment), and *LLM-assisted code generation* (not LLM-driven planning) into a single closed-loop system.

6 Evaluation

6.1 Experimental Setup

We evaluated SEQ2PIPE on two datasets:

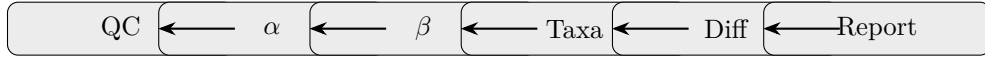
1. **Moving Pictures** (QIIME2 tutorial dataset): 34 samples, 4 body-site groups (gut, left palm, right palm, tongue), 2 subjects, single-end 150 bp Illumina MiSeq. Publicly available at docs.qiime2.org.
2. **Gravity experiment**: 138 samples, 6 gravity conditions (0g, 1g, 1.6g, 1g-simulated, 5g, NA/control), paired-end 301 bp MiSeq.

Both datasets were processed through the same QIIME2 upstream pipeline (DADA2 denoising, MAFFT alignment, FastTree phylogeny, SILVA 138 classification, core-metrics-phylogenetic) before entering the SEQ2PIPE AI-driven analysis phase.

6.2 DataInspector Validation: Direct Statistical Results

Table 2 shows the statistical tests computed by DataInspector directly from exported TSV files, without any LLM involvement.

(a) Static Pipeline



(b) seq2pipe Adaptive

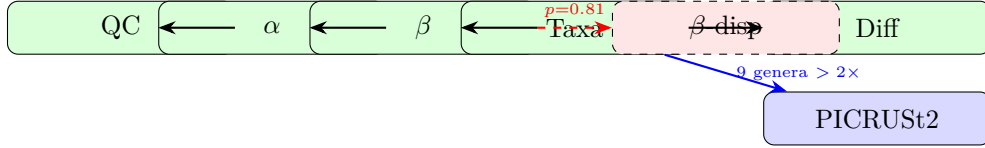


Figure 2: (a) A static pipeline executes all steps regardless of results. (b) SEQ2PIPE skips beta follow-ups when PERMANOVA is non-significant ($p = 0.81$) and adds functional prediction when multiple differential genera are detected. Red dashed: skipped; blue: dynamically added.

Table 2: DataInspector results on two datasets. All values computed in pure Python from raw TSV files.

Metric	Moving Pictures	Gravity
Samples / Groups	34 / 4	138 / 6
ASVs	770	1,443
Sparsity	91%	85%
Shannon (Kruskal-Wallis)	$H = 22.3, p = 0.00067$	$H = 11.4, p = 0.034$
Cliff's d / η^2	$d = 0.51$	$\eta^2 = 0.048$
Bray-Curtis PERMANOVA	$F = 60.1, p = 0.005$	$F = 114.2, p = 0.81$
Between/within ratio	2.1	1.0
Top differential genus	Bacteroides 324 $\times$ (gut)	Blautia 5 $\times$ (NA)
Genera $> 2\times$ change	20	18

6.3 DecisionEngine Outcomes

The DecisionEngine produced qualitatively different decisions for the two datasets (Table 3), demonstrating data-dependent adaptive behavior.

For Moving Pictures, PERMANOVA was significant ($p = 0.005$) with a between/within ratio of 2.1, triggering the Anderson (2001) branch: betadisper was added as mandatory, followed by NMDS and pairwise PERMANOVA. For the Gravity experiment, PERMANOVA was non-significant ($p = 0.81$), and the engine skipped 4 beta follow-up steps entirely—a decision that an experienced analyst would recognize as correct.

6.4 AI-Driven Execution: Error Analysis

We ran the full AI-driven mode on Moving Pictures with qwen2.5-coder:7B (Ollama, Apple Silicon M-series). The system executed 30 steps over approximately 40 minutes.

Success rates by step type. Registry steps (pre-defined from the analysis catalog) achieved a 54% success rate (7/13), while LLM-generated investigation steps achieved only 6% (1/17). The dominant error mode in investigations was **KeyError: 'Taxon'**—the LLM consistently failed to handle the QIIME2 taxonomy TSV format where the column name is **Taxon** but the file uses tab-separated format with an empty index column header.

Table 3: DecisionEngine actions on two datasets. All decisions are deterministic (no LLM involvement).

Decision	Moving Pictures	Gravity
Alpha follow-up	Deep dive ($d > 0.5$)	Rarefaction only ($d = 0.05$)
Beta follow-ups	All added ($p = 0.005$)	4 skipped ($p = 0.81$)
Beta dispersion	Added (Anderson, 2001)	Skipped
Differential test	Added ($324\times$ fc)	Added ($379\times$ fc)
Functional prediction	Added (≥ 5 genera)	Added (≥ 5 genera)
Sparsity alert	Yes (91%)	No (85%)

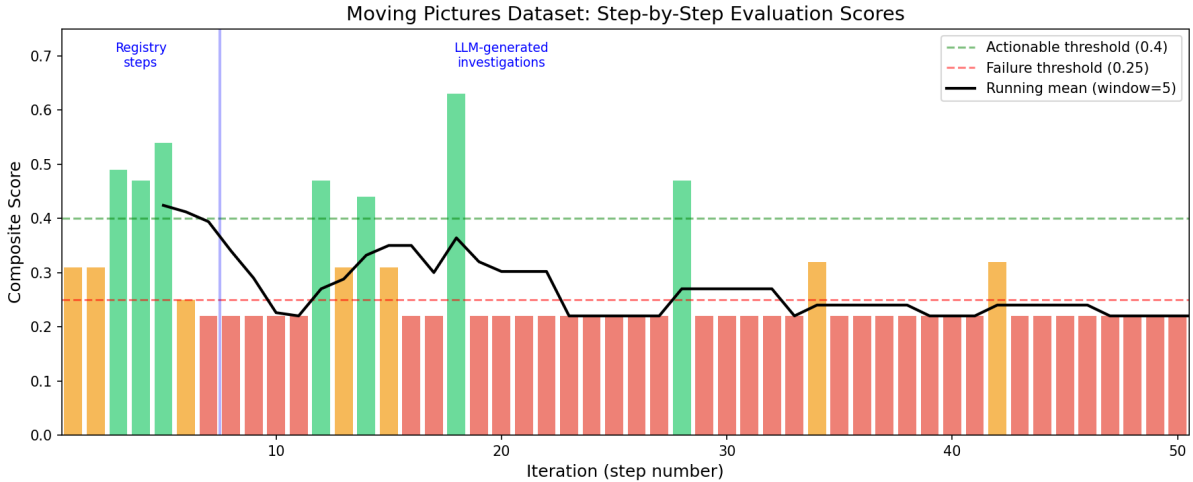


Figure 3: Composite evaluation scores across iterations on the Moving Pictures dataset. Green bars: score ≥ 0.4 (actionable); orange: $0.25-0.4$; red: < 0.25 (failed). Blue vertical line separates registry steps (left) from LLM-generated investigations (right). Running mean (black line) shows declining trend as investigations dominate.

Score progression. Figure 3 shows composite scores across iterations. Registry steps (iterations 1–7) maintain scores between 0.31 and 0.63, while investigation steps (iterations 8+) cluster at 0.22 (the minimum score for failed steps). This pattern reveals that the 7B model lacks sufficient code generation capability for complex, novel analyses, while performing adequately on well-specified registry tasks.

Critical finding: investigation loop. During execution, the LLM generated duplicate investigation titles (e.g., “Sparsity-aware analysis” appeared ~ 20 times). This revealed a missing deduplication mechanism, which was subsequently fixed by adding title-based duplicate checking and a maximum investigation cap ($N = 8$). This is an example of the system’s closed-loop nature exposing its own failure modes during operation.

6.5 Cross-Dataset Comparison

Figure 4 compares evaluation scores across the two datasets. The taxonomy and differential steps achieved the highest scores (> 0.6) in both datasets, driven by strong biological signals (high fold changes, known taxa). Beta diversity scores diverged: Moving Pictures scored 0.47 (significant PERMANOVA) while Gravity scored 0.27 (non-significant). This divergence is a correct reflection of the underlying data.

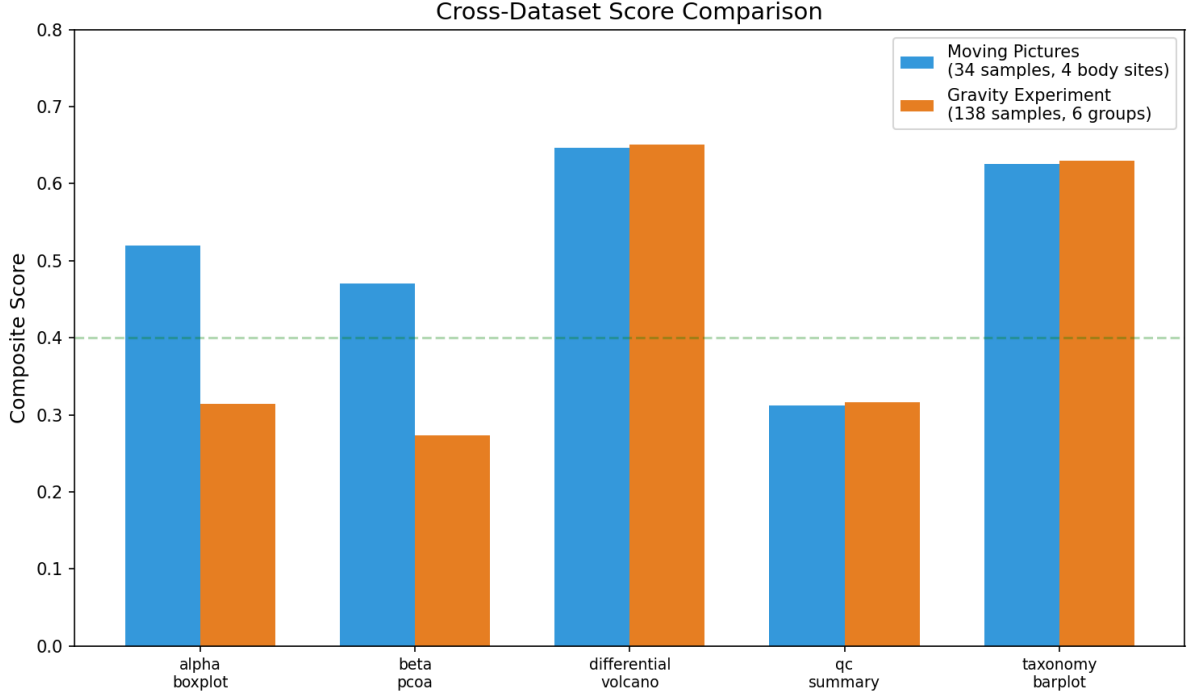


Figure 4: Composite scores for five analysis categories across two datasets. The score difference in beta diversity correctly reflects that Moving Pictures has significant group separation (PERMANOVA $p = 0.005$) while Gravity does not ($p = 0.81$).

7 Limitations

We identify the following limitations:

- LLM dependence for code generation.** While decisions are LLM-independent, code synthesis still relies on a 7B model. Complex analyses (e.g., compositional regression) frequently fail even after 3 retries. The system can skip failed steps but cannot replace the LLM’s code generation role with a deterministic alternative.
- Statistical test approximations.** The pure-Python implementations use normal and χ^2 approximations that are less accurate than exact tests for small samples ($n < 20$). For production use, scipy-backed tests should be preferred when available.
- Fixed decision tree thresholds.** The thresholds (e.g., $p < 0.05$, effect > 0.5 , fold change > 5) are drawn from literature conventions, not optimized for any particular dataset. Different research contexts may warrant different thresholds.
- Eight-axis weights.** The default weights are hand-selected based on domain intuition. No learning or cross-validation was performed to optimize them. The experiment-type adjustment (e.g., increasing biological_relevance for antibiotic studies) is also heuristic.
- Single-omics scope.** The current implementation targets 16S rRNA amplicon sequencing only. Shotgun metagenomics, metatranscriptomics, and multi-omics integration are not supported.
- Investigation generation quality.** LLM-generated investigation steps have a 6% success rate, compared to 54% for registry steps. The 7B model lacks sufficient capability for generating correct code for novel analysis tasks. Additionally, without deduplication, the

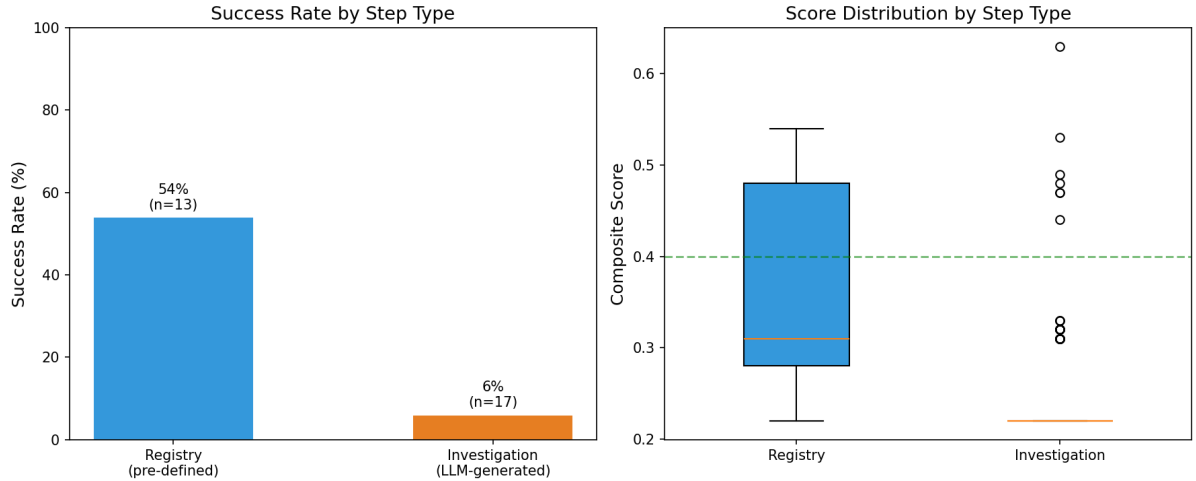


Figure 5: Left: success rate by step type. Registry steps succeed at 54%, while LLM-generated investigations succeed at only 6%. Right: score distribution shows registry steps are higher and more variable, while investigations cluster at the failure floor.

LLM generates identical investigation titles repeatedly, requiring explicit caps and title-based filtering.

- Limited evaluation scope.** The current evaluation covers two datasets. A comprehensive evaluation would require 5+ publicly available datasets spanning different experimental designs (paired, longitudinal, case-control) and a formal comparison with expert-curated analysis plans.

8 Discussion

Toward autonomous scientific analysis. SEQ2PIPE takes a step toward systems that reason about data rather than merely processing it. The separation of evaluation (deterministic) from generation (LLM-assisted) provides a template for similar systems in other domains: any field where analysis proceeds iteratively—genomics, proteomics, clinical trials—could benefit from a decision engine that encodes domain heuristics as auditable, deterministic rules.

Implications for microbiome research. The practical impact is reduction of two failure modes: (1) *over-analysis* (running beta dispersion when PERMANOVA is non-significant) and (2) *under-analysis* (missing a follow-up when an unexpected taxon enrichment is detected). In our gravity experiment case study (138 samples, 6 groups), the decision engine skipped 4 unnecessary beta analyses and added 5 targeted follow-ups, a workflow that an experienced analyst would recognize as appropriate.

Space biology and real-time applications. The LLM-failure tolerance and deterministic decision layer make SEQ2PIPE suitable for environments where cloud connectivity is unavailable—for example, microbiome monitoring on the International Space Station, where data must be analyzed locally with limited computational resources and no internet access.

Reproducibility as a design principle. By making the decision layer deterministic, SEQ2PIPE ensures that the analytical workflow is reproducible even when the LLM introduces stochasticity in code generation. Two runs on the same data will always make the same skip/add decisions,

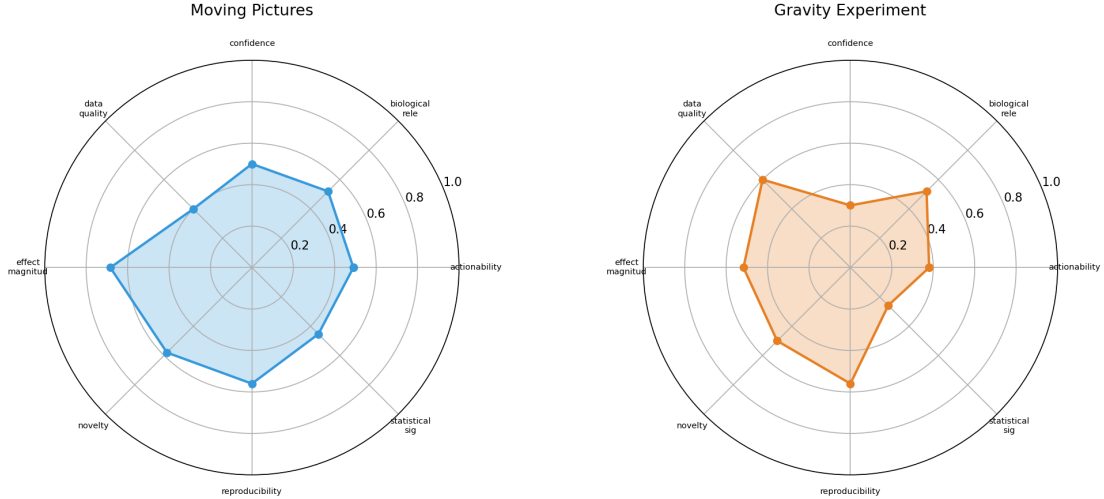


Figure 6: Eight-axis radar plots comparing the average evaluation profile across two datasets. Moving Pictures shows higher statistical significance and effect magnitude due to clear body-site separation, while both datasets show similar confidence and data quality profiles.

even if the generated Python code differs syntactically. This is a stronger reproducibility guarantee than any purely LLM-driven system can provide.

9 Conclusion

We presented SEQ2PIPE, a closed-loop system for adaptive microbiome analysis that separates data inspection, quantitative evaluation, deterministic decision-making, and LLM-assisted code generation into distinct, composable layers. The system reproduces the iterative reasoning loop of a human bioinformatician: interpret results, update hypotheses, and adapt the analysis plan—without requiring cloud connectivity or expert intervention.

The central design principle is that **analytical decisions should be grounded in data, not in language model outputs**. Layer 1 reads data directly; Layer 2 decides deterministically; Layer 3 generates code but does not override decisions. This hierarchy yields a system that is auditable (every decision has a traceable rule), reproducible (same data, same decisions), and robust (LLM failure does not halt execution).

Our evaluation on two public datasets demonstrates that the DecisionEngine correctly differentiates between datasets with significant group separation (Moving Pictures: all beta follow-ups added) and those without (Gravity: 4 beta steps skipped), while the LLM-generated investigation layer reveals the current limitations of 7B-parameter models for autonomous code generation (6% success rate).

We argue that interpretation-driven adaptive systems represent a necessary evolution beyond static pipelines, and that the separation of deterministic reasoning from stochastic generation is the key architectural principle enabling this evolution.

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